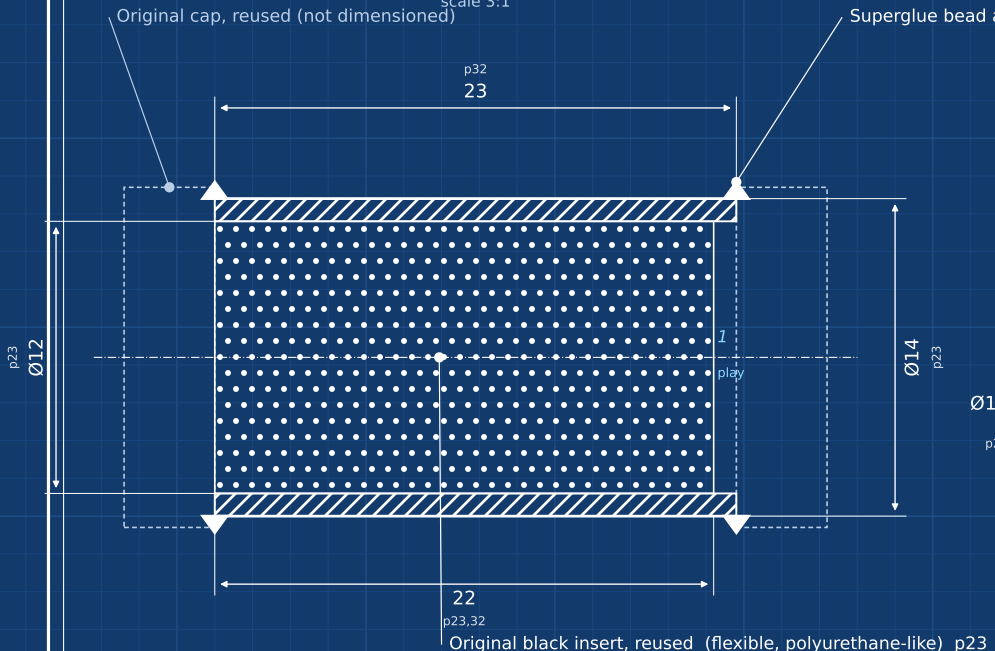
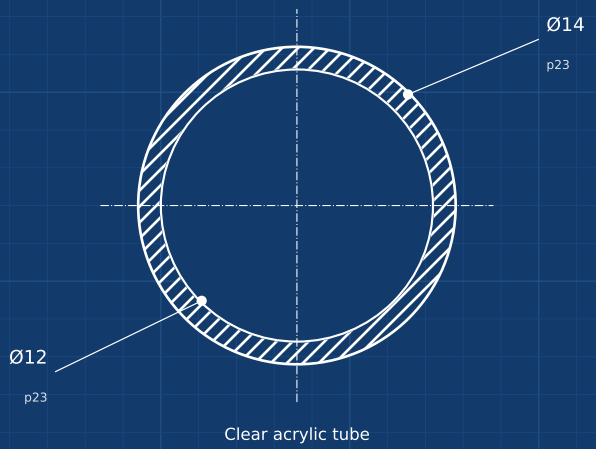
SECTION A–A · ASSEMBLED TUBE

scale 3:1



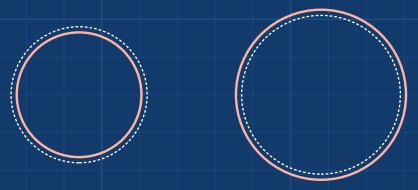
END VIEW · TUBE

scale 3:1



SIZES THAT DID NOT WORK

scale 1.5:1 · dashed = size that worked



Ø11 ID

p12,23

ID too small for the inserts (need Ø12)

Ø15 OD

p23

OD too big for the lid housing (Ø14 fits)

PROCEDURE

1. Free the caps: clamp the middle of each stock tube with locking pliers and squeeze it oval. The glue seal breaks and the caps pop off. The clear tubes are sacrificed; caps and inserts survive.
   Source: post #23 (gbrich)
2. Buy clear acrylic tube Ø14 OD × Ø12 ID. gbrich bought 305 mm, enough for the four tubes plus spares.
   Source: post #23 (gbrich)
3. Cut to 23 mm: the insert is 22 mm long without its nub, leaving 1 mm of play. A 60T fine-cut mitre saw gave square ends; a junior hacksaw struggled.
   Source: post #23 (gbrich), post #32 (gbrich)
4. Glue a cap to one end, slide the insert in from the other end, then glue that end into the lid. The rebuilt tubes no longer twist-lock.
   Source: post #32 (gbrich)

NOTES

1. The black inserts and caps can be painted before assembly (gbrich: copper inserts, chrome caps).
   Source: post #23 (gbrich), post #32 (gbrich)
2. mcspain also rebuilt his tubes from acrylic and discarded about half of his cuts. Allow spare tube.
   Source: post #81 (mcspain)
3. Cap geometry is not documented and is drawn schematically.

HASBRO 1984 PROTON PACK MODDING

GBFans.com community guide · drawn from forum-reported measurements

CRT EMITTER TUBE REBUILD

SHEET **S1 (1 of 7)**

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #12, #23, #32, #81

SCALE 3:1

Dimensions in mm unless noted

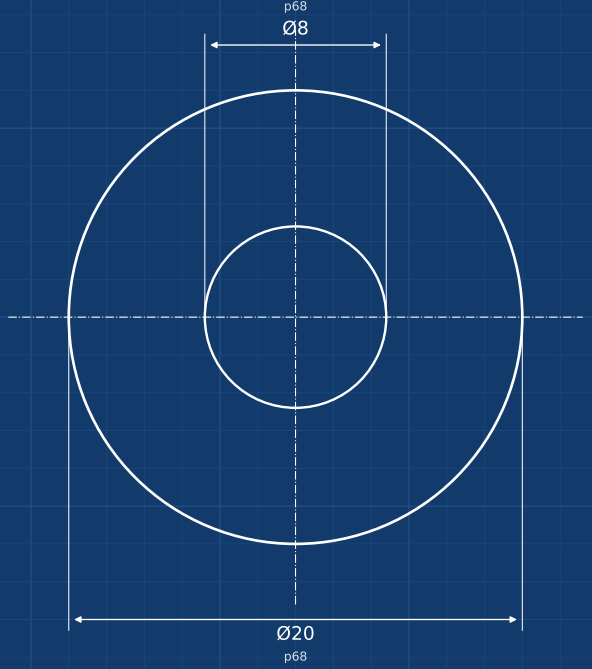
v0.2 · 2026-11-02

Drawn: ryan-brumback · not a Hasbro drawing

Builder-reported values from GBFans.com. Check against your own pack before cutting or drilling. Attribution and forum photos: docs/RENDERINGS.md

PLAN VIEW

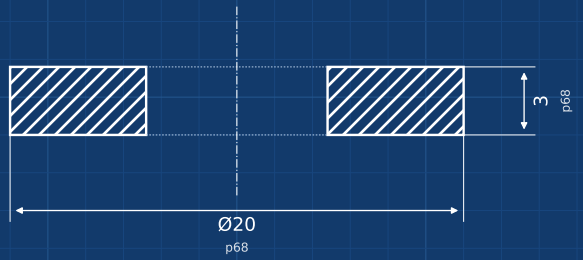
scale 3:1



SECTION B–B

scale 3:1

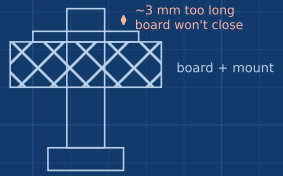
Printed in PLA by gbrich; rubber washers or other spacers also work p68



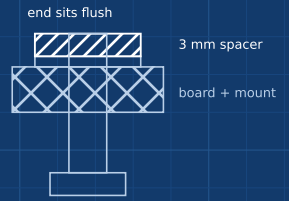
WHY IT'S NEEDED

SCHEMATIC — NOT TO SCALE

BEFORE



AFTER



NOTES

1. With Redman's 1984 ALICE frame mounting kit, the bolts were about 3 mm too long for the aluminum motherboard to close flush.
   Source: post #68 (gbrich)
2. gbrich considered grinding the bolt ends, reversing the bolts, or spacers. He printed these spacers and fitted them under the outer washers.
   Source: post #68 (gbrich)
3. The stack-up diagram shows the principle only. Hardware shapes are not documented.

HASBRO 1984 PROTON PACK MODDING

GBFans.com community guide · drawn from forum-reported measurements

ALICE FRAME MOUNTING SPACER

SHEET **S2 (2 of 7)**

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #68

SCALE 3:1

Dimensions in mm unless noted

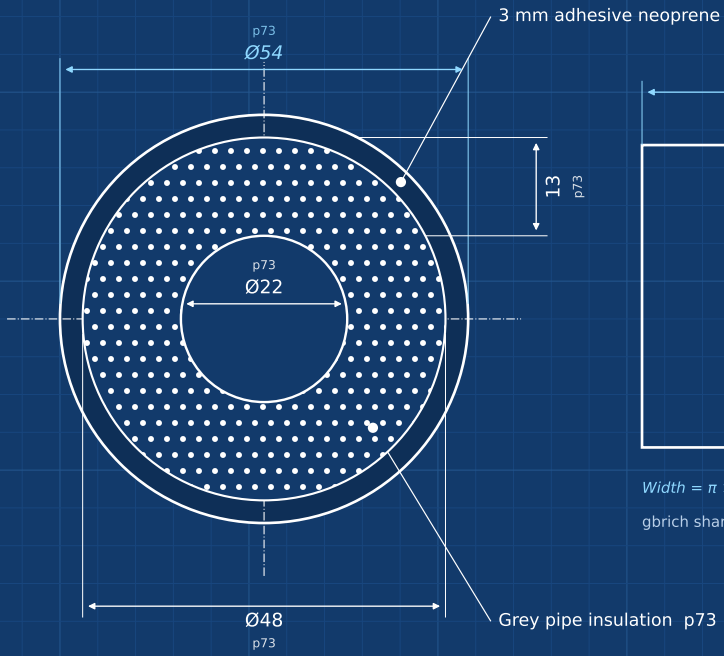
v0.2 · 2026-11-02

Drawn: ryan-brumback · not a Hasbro drawing

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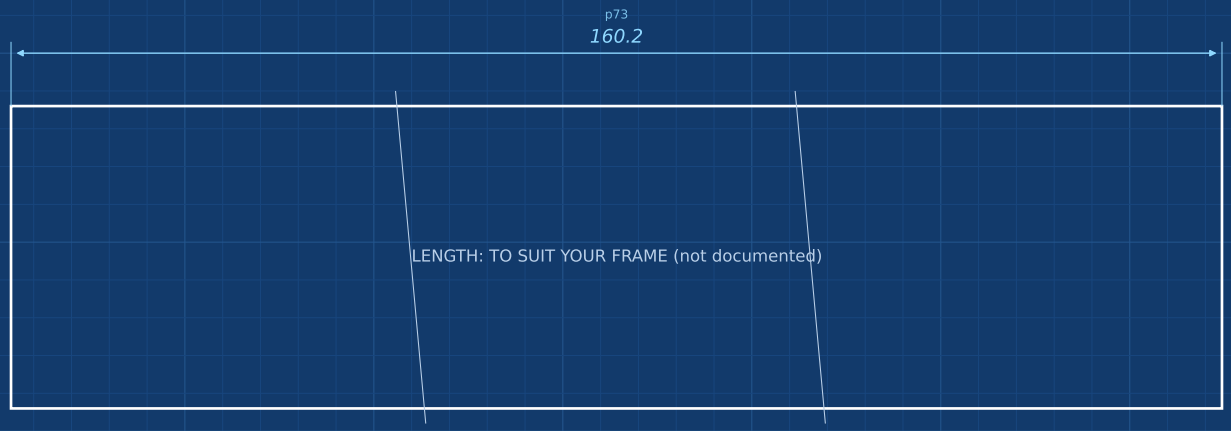
SECTION · NECK PAD

scale 1:1



COVERING · FLAT PATTERN

scale 1:1 (print at 100%)



Width = π × (48 + 3) = 160.2 mm, measured at the sheet's mid-thickness. Add a small overlap if you butt-join with adhesive.

gbrich shared his own measurement template for any tube diameter (see RENDERINGS.md).

NOTES

1. Core: grey pipe insulation, 22 mm ID, 13 mm wall (48 mm OD). Covering: 3 mm black adhesive neoprene/foam sheet cut to fit.
   Source: post #73 (gbrich)
2. After a full day at MCM Comic Con London the top nut of the ALICE frame still dug in through two layers of 3 mm padding. Plan for more.
   Source: post #88 (gbrich)
3. 3 mm neoprene left marks on the pack; gbrich is considering black felt.
   Source: post #91 (gbrich)

HASBRO 1984 PROTON PACK MODDING

GBFans.com community guide · drawn from forum-reported measurements

ALICE FRAME NECK PAD

SHEET **S3 (3 of 7)**

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #73

SCALE 1:1

Dimensions in mm unless noted

v0.2 · 2026-11-02

Drawn: ryan-brumback · not a Hasbro drawing

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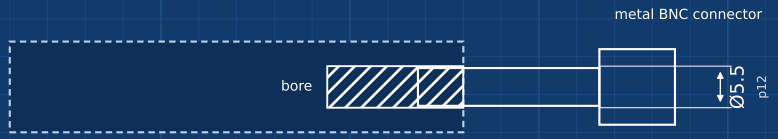
CABLE END · BEFORE AND AFTER

SCHEMATIC — NOT TO SCALE

STOCK



MODIFIED



OPTIONS

1. Drill Ø5.5 into the cable stub: the connector pushes in and holds by friction, no glue needed.
   Source: post #12 (gbrich)
2. Drill Ø6 if that's the bit you have, and glue the connector in.
   Source: post #12 (gbrich)
3. Slice the molded ends off with a scalpel blade first. gbrich bought a pack of 10 connectors for about £5.
   Source: post #12 (gbrich)
4. Bore depth, cable diameter and connector sizes are not documented: proportions here are illustrative.

Builder-reported values from GBFans.com. Check against your own pack before cutting or drilling. Attribution and forum photos: docs/RENDERINGS.md

HASBRO 1984 PROTON PACK MODDING

GBFans.com community guide · drawn from forum-reported measurements

BNC CABLE ENDS

SHEET **S4 (4 of 7)**

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #12

SCALE NOT TO SCALE

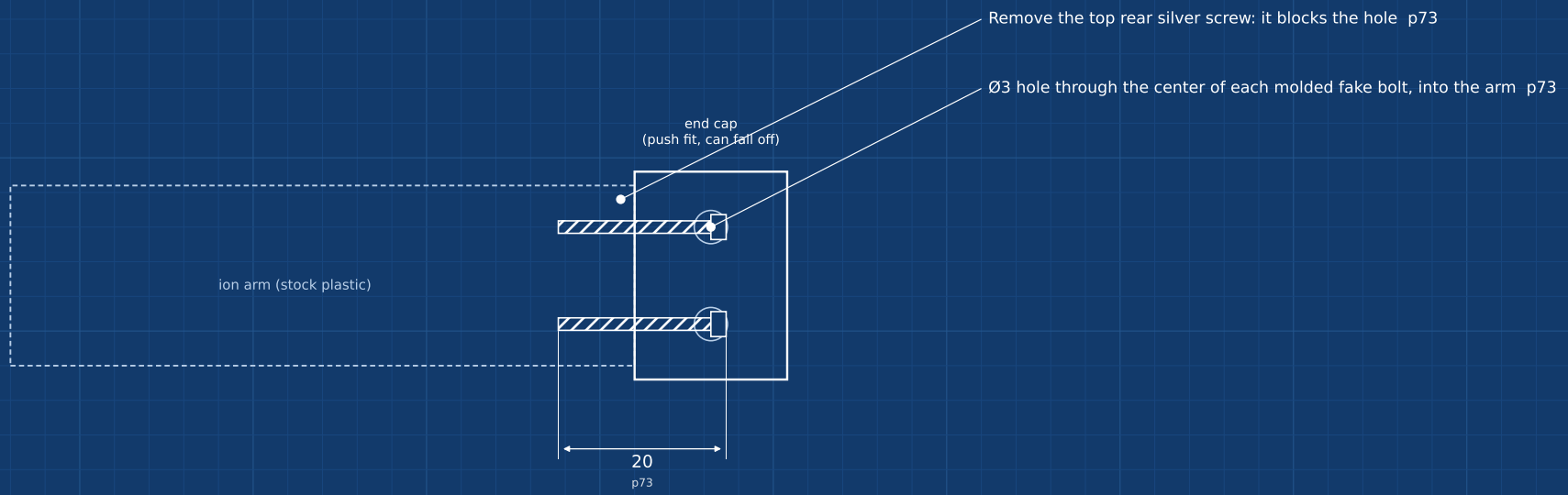
Dimensions in mm unless noted

v0.2 · 2026-11-02

Drawn: ryan-brumback · not a Hasbro drawing

ION ARM END CAP · SIDE VIEW

SCHEMATIC — NOT TO SCALE



PROCEDURE

1. Drill two Ø3 mm holes through the centers of the two molded fake bolts on the end cap, into the ion arm.
   Source: post #73 (gbrich)
2. Fit two M3 × 20 mm bolts. Remove the top rear silver screw first.
   Source: post #73 (gbrich)
3. Alternative: an aluminum ion arm (GBFans shop).
   Source: post #73 (gbrich)
4. Positions and proportions are illustrative; only the hole and bolt sizes are documented.

HASBRO 1984 PROTON PACK MODDING

GBFans.com community guide · drawn from forum-reported measurements

ION ARM END CAP PINNING

SHEET **S5 (5 of 7)**

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #73

SCALE NOT TO SCALE

Dimensions in mm unless noted

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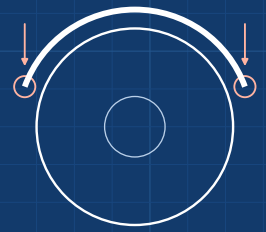
Drawn: ryan-brumback · not a Hasbro drawing

Builder-reported values from GBFans.com. Check against your own pack before cutting or drilling. Attribution and forum photos: docs/RENDERINGS.md

HGA REMOVAL · nomake-wan's non-destructive method

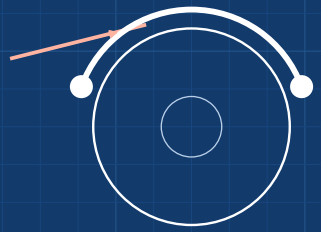
SCHEMATIC — NOT TO SCALE

1 Remove the 2 bracket screws



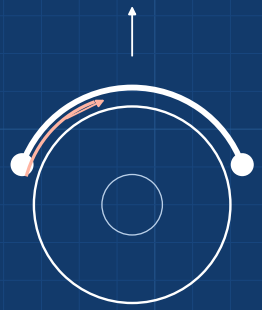
screws out

2 Blade under bracket, sharp edge up



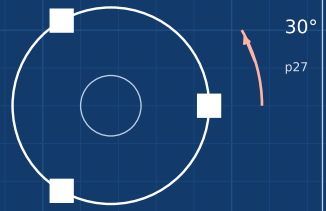
push until it bottoms out

3 Draw blade up the curve; lift bracket



cuts the superglue (expect pops)

4 Rotate 30°, slide out



tabs line up with the slots

NOTES

1. Use a thin safety blade (nomake-wan used the one from a Tamiya basic tool set). Worked on both a HasLab and a 1984 pack with no damage to the HGA or bracket.
   Source: post #27 (nomake-wan)
2. Once unscrewed, a pry tool can also pop the glue off the band around the HGA.
   Source: post #21 (spooky)
3. Remove the HGA before cutting off the fake bolts: the cut is much cleaner.
   Source: post #29 (gbrich)
4. mcspain found the twist lock holds the HGA firmly even without the bracket.
   Source: post #81 (mcspain)
5. The HGA, bracket and tab positions are drawn schematically. See nomake-wan's photos for the blade position.

HASBRO 1984 PROTON PACK MODDING

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HGA REMOVAL (NON-DESTRUCTIVE)

SHEET **S6 (6 of 7)**

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #27, #81

SCALE NOT TO SCALE

Dimensions in mm unless noted

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Drawn: ryan-brumback · not a Hasbro drawing

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HOLE SIZES AT TRUE SIZE

scale 1:1 — print at 100% and check the 50 mm bar



M2.5

M2.5 screws: cake lid window

post #49, post #68



Ø4.5 mm

Wand wing, Mack's banjo screw (4–4.5)

post #79



M6

Real bolts on the HGA (originals 1/4-20)

post #20, post #22, post #30



Ø3 mm

C&A ribbon cable clamp: 2 new holes

post #73



Ø5 mm

Aluminum board over speaker: 7 holes

post #68



Ø6.5 mm

Dummy toggle switches in cyclotron

post #85



Ø3 mm

Ion arm end cap: 2 holes for M3 × 20

post #73



Ø5.5 mm

BNC into cable stub: friction fit

post #12



Ø7.94 mm (5/16")

Redman shell bolts, per instructions (5/5/16")

post #68



M3

M3 bolts + nuts in place of rivets

post #73



Ø6 mm

BNC into cable stub: with glue

post #12



Ø9 mm

Brass elbows / Legris (stepped from 8)

post #4

Pilot holes for the M6 bolts: slightly smaller than the bolt shank so the bolt threads into the plastic (exact bit not recorded, post #22).

Cosmetic Redman shell bolts: start small and enlarge until the bolt threads in snugly; only one bolt per bracket is structural (post #68).

50 mm check bar



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HASBRO 1984 PROTON PACK MODDING

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DRILL AND HARDWARE CHART (PRINT AT 100%)

SHEET

S7 (7 of 7)

SOURCED ***DERIVED*** **- - SCHEMATIC** pN = forum post number

Source: thread 219236, posts #4, #12, #20, #22, #30, #49, #68, #73, #79, #85

SCALE 1:1

Dimensions in mm unless noted

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Drawn: ryan-brumback · not a Hasbro drawing